

CSCN8020 Assignment 3 Submission

Deep Q-Network Control of the Unitree G1 Left Elbow

Student full name	Haibo Yuan
Student ID	9010929
Assignment title	CSCN8020 Assignment 3 - Unitree G1 DQN Left Elbow Control

Project Summary

This project implements a student-written PyTorch Deep Q-Network for the approved Unitree G1 left elbow Gymnasium environment. The DQN maps the elbow angle, velocity, target angle, and angle error to three discrete action values: decrease, hold, or increase the controller target. Two epsilon-decay settings were trained headlessly on CPU and evaluated greedily across four benchmark goals. The selected checkpoint reached 20/20 successful final evaluation episodes and was compared with the provided rule-based policy using saved metrics, plots, a technical report, and a rendered MuJoCo demonstration video.

GitHub Repository URL

https://github.com/hyuan0929/CSCN8020_Assignment3

Cloneable .git URL

https://github.com/hyuan0929/CSCN8020_Assignment3.git

Rendered Evaluation Video

https://drive.google.com/file/d/1Av66sldDlvqDU2o0uGR-4OtaGb_OI2Q3/view?usp=drive_link

Submission Notes

The repository contains the completed notebook, student-written DQN source files, saved checkpoint, metrics, plots, README commands, and the full technical report. The rendered video is provided as an accessible Google Drive link because the Brightspace dropbox allows only one PDF file.